

Sprint 2 Technical Report

Real-Time Customer Support Ticket CRM (TicketBloom)

Submitted By:

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1 Project Overview

The Real-Time Customer Support Ticket CRM, branded as TicketBloom, is a full-stack web application designed to manage customer support operations efficiently. The system allows users to raise support tickets while enabling support agents to view, manage, and update ticket statuses in real time.

Sprint 1 focused on building the foundation of the system, including user authentication using JWT, password hashing using bcrypt, and MongoDB integration. Sprint 2 builds upon this foundation and introduces the complete ticket lifecycle system with Role-Based Access Control (RBAC) and an interactive dashboard interface.

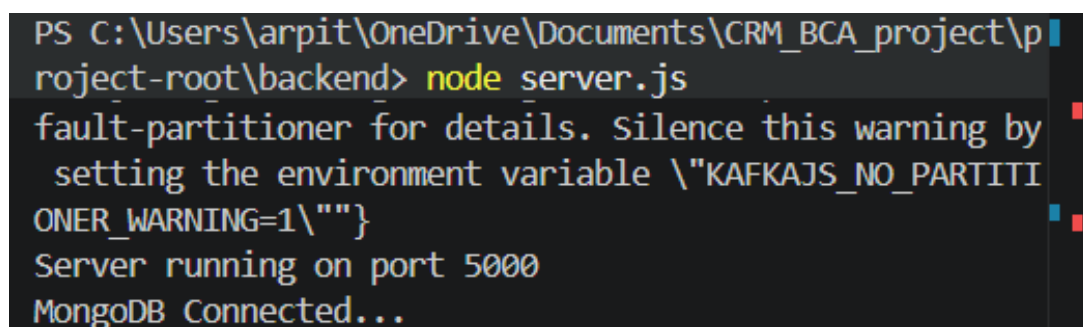
2 Sprint 2 Objectives

The primary objective of Sprint 2 was to implement the ticket management system. The key goals included:

- Designing the Ticket schema using Mongoose
- Implementing ticket creation API
- Developing role-based ticket retrieval (RBAC)
- Implementing ticket status update functionality
- Building a responsive dashboard interface

3 Backend Execution

The backend server was developed using Node.js and Express.js. MongoDB was integrated using Mongoose to store and manage application data.

A terminal window with a dark background and light-colored text. The command prompt shows the user is in the directory 'C:\Users\arpit\OneDrive\Documents\CRM_BCA_project\project-root\backend'. The command 'node server.js' has been executed. The output shows a warning about the 'fault-partitioner' for KafkaJS, followed by instructions to silence it by setting an environment variable. Below the warning, it says 'Server running on port 5000' and 'MongoDB Connected...'.

```
PS C:\Users\arpit\OneDrive\Documents\CRM_BCA_project\project-root\backend> node server.js
fault-partitioner for details. Silence this warning by
  setting the environment variable \"KAFKAJS_NO_PARTITIONER_WARNING=1\"}
Server running on port 5000
MongoDB Connected...
```

Figure 1: Backend Server Running with Successful MongoDB Connection

The terminal output confirms that the server is running successfully and connected to MongoDB.

4 Database Implementation

MongoDB is used as the database for storing ticket information. A structured schema was created to maintain ticket data integrity.

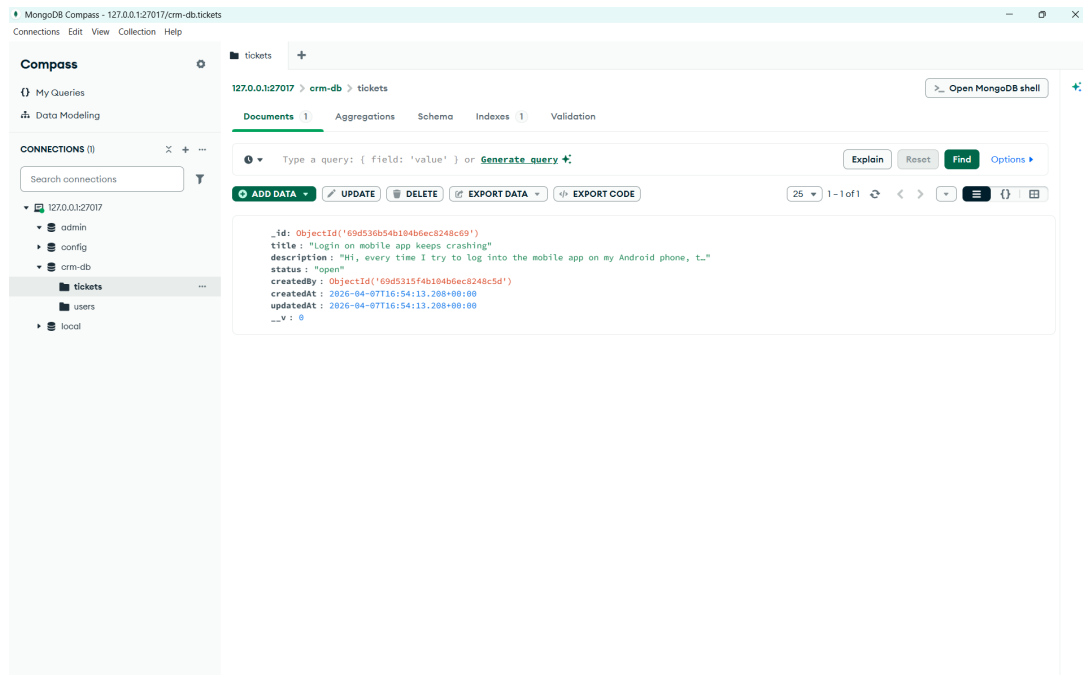


Figure 2: MongoDB Compass — Ticket Collection with Stored Ticket Document

The database contains fields such as title, description, status, and author reference, validating successful data storage.

5 Authentication System

The login page enables users to securely access the system using JWT-based authentication.

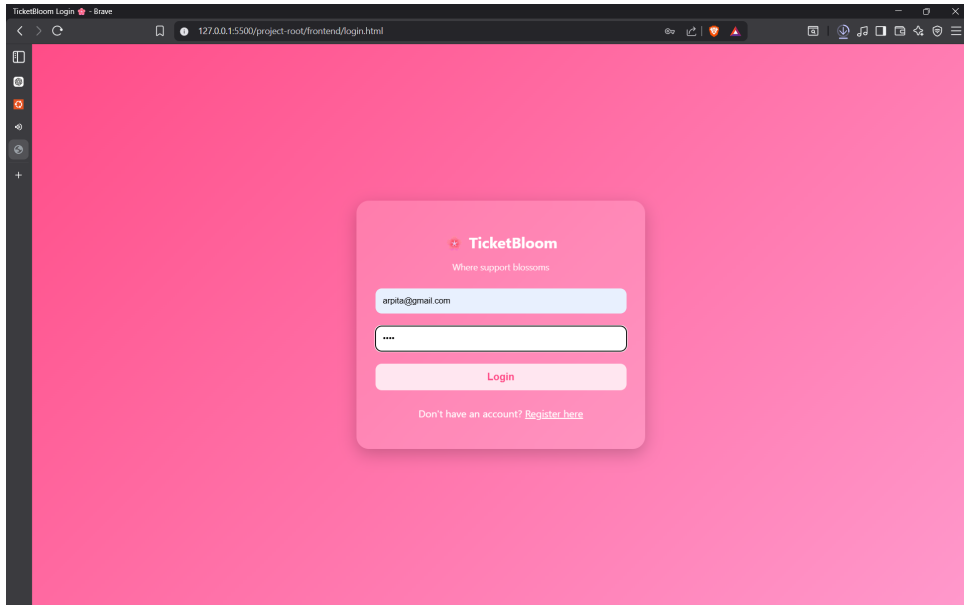


Figure 3: TicketBloom Login Page (Authentication Interface)

This ensures secure and role-based access to the application.

6 Dashboard and Ticket Management

The dashboard provides a user-friendly interface for managing tickets. It displays ticket statistics and allows users to perform ticket-related operations.

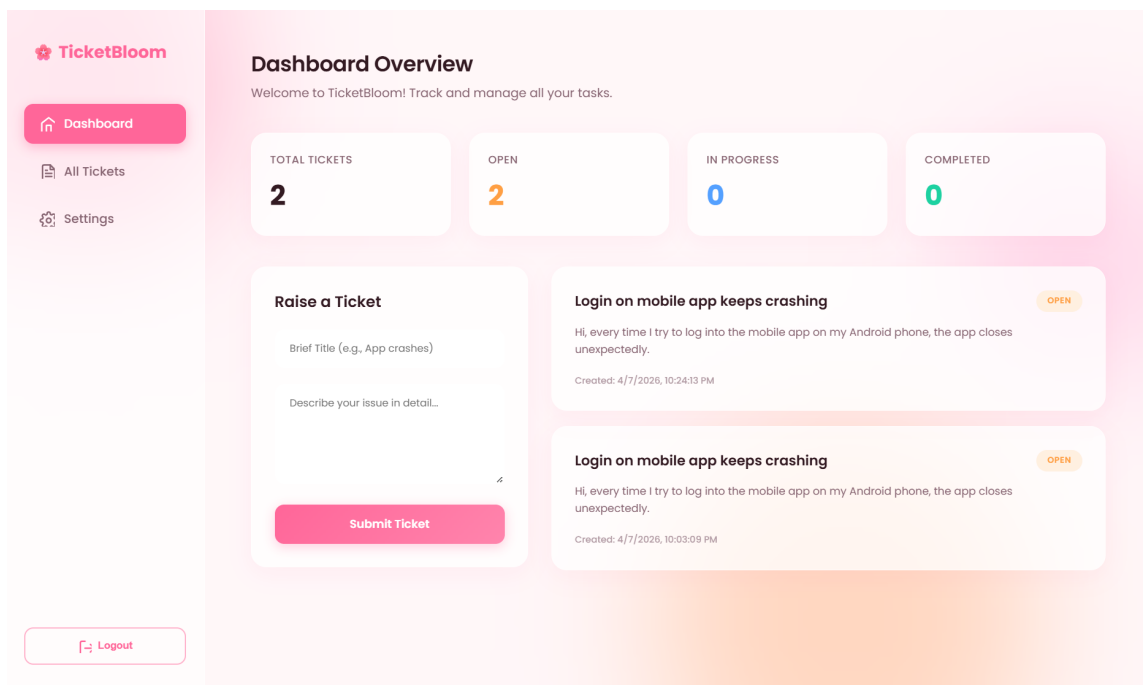


Figure 4: Dashboard Overview — Ticket Management Interface with Stats and Ticket List

The dashboard successfully demonstrates:

- Ticket creation functionality
- Ticket listing and visualization
- Status tracking system
- Role-Based Access Control (RBAC)

7 Conclusion

Sprint 2 successfully implemented the complete ticket management system. The backend APIs, database integration, and frontend dashboard are fully functional. Role-Based Access Control ensures that users and agents have appropriate access to system data.

The system is now prepared for further enhancement with Apache Kafka integration in Sprint 3, which will enable real-time event-driven communication.